Visualization of Single Perceptron

$$X = [x_1; x_2; \ldots; x_n] \rightarrow X = [1; x_1; x_2; \ldots; x_n]$$

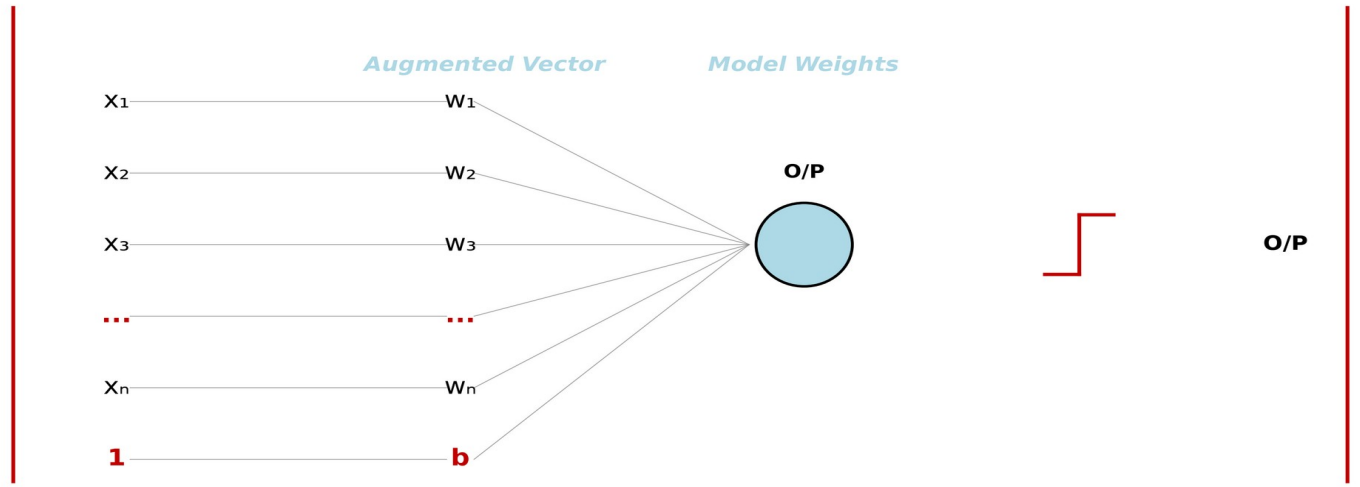
Augmented Vector

$$W = [b; w_1; w_2; \ldots; w_n]$$

Model Weights

$$d(X) = W^T X = w_1 x_1 + w_2 x_2 + \ldots + w_n x_n + b \cdot 1$$

- $>0 \Rightarrow$ ***Class I***
- $<0 \Rightarrow$ ***Class II***
- $=0 \Rightarrow$ *Undefined Class*



Binary Classification (2 Classes)